

Semantic Tokenization Specification

Universal Bridge (URIB)

Meaning-Preserving Identity, Continuity, and Cross-Rail Interpretation

Universal Bridge Framework · Semantic Layer Specification

Semantic Tokenization Specification Universal Bridge (URIB)

Normative reference for semantic graph construction, entity & relationship taxonomy, invariant enforcement, meaning-preservation, and semantic commitment contracts.

Document ID

URIB-SPEC-001

Version

v1.0.0

Status

✓ Authoritative

Issued

July 2026

Discipline

Universal Bridge — Semantic Layer

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Usage & Distribution Notice:

This document is the authoritative normative specification for URIB-conformant runtime implementations. It supersedes all prior informal specifications and Base44 platform implementation notes. Distribution is permitted to licensed URIB implementors. Normative keywords (

MUST

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MUST NOT

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SHALL

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SHOULD

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MAY

) are used per

RFC 2119

. Any implementation deviating from MUST-level requirements is non-conformant.

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Abstract

This document defines the normative Semantic Tokenization model for URIB systems. It is the authoritative specification for semantic graph construction, entity and relationship taxonomy, invariant enforcement, meaning-preservation rules, and semantic commitment contracts. This specification is intended for engineers implementing URIB-conformant runtimes and is immediately actionable as a design and validation reference.

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1. Introduction & Purpose

1.1 What is URIB?

The **Universal Bridge (URIB)** is a framework architecture that unifies heterogeneous data systems across organizational, technical, and domain boundaries through a shared semantic layer. Rather than forcing point-to-point data transformations — which proliferate into $O(n^2)$ integration complexity — URIB defines a single canonical meaning surface: the **Semantic Graph**. Every participating system speaks to the graph. The graph speaks meaning.

URIB targets the class of integration problems where raw data connectivity is insufficient: scenarios where two systems can exchange bytes but cannot exchange *understanding*. It is particularly applicable to multi-domain enterprise architectures, cross-organizational data federations, AI reasoning pipelines, and event-driven systems where event semantics must survive routing and processing boundaries.

1.2 What is Semantic Tokenization?

Semantic Tokenization is the foundational operation of the URIB semantic layer. It is the process of transforming raw data primitives — strings, scalar values, event records, object references, binary payloads — into **typed, graph-anchored semantic tokens** that carry explicit meaning, provenance, and relationship context.

A semantic token is not merely a labelled data record. It is a graph node with a declared type from the URIB Entity Taxonomy, a confidence score, a provenance chain, a mutability class, and a set of typed relationships to other tokens. Every semantic token is a first-class citizen of the semantic graph — it carries meaning not just through its payload, but through its position, connections, and declared commitments.

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Semantic Tokenization preserves intent through every system boundary crossing. When a token travels from a Commerce Domain to an Analytics Domain, its meaning travels with it — enforced by the Meaning-Preservation Rules in Section 7 and validated by the Semantic Invariants in Section 6.

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1.3 Why It Matters

Without semantic tokenization, meaning degrades at every integration boundary. A field named `customer_id` in one system becomes `cust_ref` in another and `account_number` in a third. The string value survives. The identity contract does not. An event timestamped in UTC loses its temporal ordering semantics when received by a system that assumes local time. A confidence score of 0.87 in a source ML model becomes an untyped float in the receiving ETL pipeline — its calibration context, decay rules, and floor constraints silently discarded.

URIB makes meaning a **first-class citizen**. The semantic layer is not a documentation layer — it is a runtime enforcement layer. Invariants are checked. Contracts are machine-evaluated. Violations are raised, not silently ignored.

1.4 Scope of this Specification

This specification governs:

- All semantic graph construction and mutation operations in URIB-conformant runtimes
- The normative Entity Type Taxonomy (Section 4) and Relationship Type Taxonomy (Section 5)
- All Semantic Invariants (Section 6) that **MUST** be enforced at runtime
- All Meaning-Preservation Rules (Section 7) that govern semantic fidelity across operations
- Semantic Commitment Contracts (Section 8) and their machine-evaluable lifecycle
- Cross-domain Bridge Operations and DomainBridge record requirements
- Conformance requirements for Level 1, Level 2, and Level 3 implementations (Section 10)

This specification does not govern physical transport protocols, serialization wire formats, or runtime orchestration scheduling. Those concerns are addressed in companion URIB specifications (URIB-TRANSPORT-001, URIB-RUNTIME-001).

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1.5 Normative Keywords

The key words **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY**, and **OPTIONAL** in this document are to be interpreted as described in **RFC 2119**. All **MUST**-level requirements constitute conformance gates for Level 1 conformance and above.

2. Conceptual Foundations

2.1 The Meaning Degradation Problem

Every data integration boundary is a potential meaning loss event. Conventional integration architectures treat data as opaque payloads to be transported and transformed. The semantic content of a datum — its type contract, its provenance, its relationship to other data, its temporal validity, its confidence bounds — is rarely encoded in the data itself and is almost never enforced at the boundary.

Three primary degradation mechanisms are identified:

- **Syntactic Collapse:** A semantically typed value is reduced to its syntactic representation (e.g., an enumerated status code serialized as a plain integer, losing its domain membership contract).
- **Context Loss:** A value that was meaningful only within a specific interpretive frame (a fiscal-year-relative date, a locale-specific currency amount) crosses a boundary without its framing context, producing silent misinterpretation.
- **Impedance Mismatch:** The source system's conceptual model has no direct correspondent in the target system, forcing a lossy approximation. Without explicit declaration, this approximation is invisible.

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URIB addresses all three through the Semantic Token model: syntactic collapse is prevented by typed payloads; context loss is prevented by CONTEXT entities and SCOPES relationships; impedance mismatches are made explicit by DomainBridge records and the SC-CROSS-DOMAIN-FIDELITY contract.

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2.2 Semantic Tokens vs. Data Tokens — Comparison

Dimension	Raw Data Token	URIB Semantic Token
Representation	Opaque bytes / untyped string	Typed node in the semantic graph
Type contract	None — inferred from field name conventions	Explicit SemanticType from normative taxonomy
Relationships	Implicit (foreign key, nesting) or none	Explicit typed edges with directionality and weight
Provenance	Rarely tracked; usually lost at first transformation	Mandatory Provenance field; DERIVES_FROM chain enforced
Confidence	Not encoded	Mandatory float [0.0–1.0]; propagated with decay rules
Mutability	Uncontrolled; any system may overwrite	Declared MutabilityClass: Immutable / Mutable / Ephemeral
Temporal validity	No validity window; implicit "current"	Explicit temporal scope (valid_from / valid_to)
Boundary behavior	Silent transformation; semantics not preserved	DomainBridge record; meaning-preservation contract required
Invariant enforcement	None	12 normative invariants checked at write time
Conflict handling	Last-write-wins (silent overwrite)	CONTRADICTS edges; mandatory conflict resolution procedure

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2.3 The Semantic Graph Mental Model

Meaning in URIB lives in a directed graph. Every **entity** (concept, artifact, agent, event, etc.) is a node. Every **relationship** between entities is a typed, directed edge. The totality of this graph — spanning all domains managed by a URIB runtime — is the **truth surface** for the system.

This mental model carries three critical implications:

1. **No isolated datum is meaningful.** A token's meaning is partly constituted by its edges. A METRIC node with value 149.99 means nothing without knowing what it MEASURES and in what CONTEXT it was recorded.
2. **Graph position is semantic.** An AGENT node that GOVERNS a domain has different semantic status than an AGENT node that merely INHABITS a ROLE in it. The topology carries meaning.
3. **The graph is the authority.** When graph state and source system state diverge, the graph's meaning-preservation rules and invariants determine which is correct — not the raw source data.

2.4 Primitive Taxonomy

All entity types in the URIB taxonomy derive from four primitive classes. These are not themselves instantiable entity types but are the conceptual root categories from which all types inherit their fundamental behavioral contracts:

Primitive Class	Description	Key Behavioral Contract	Derived Entity Types
Atomic	Scalar values: indivisible data points	Payload is a single typed scalar; no internal structure	METRIC, SIGNAL, REFERENCE
Composite	Structured aggregates of other entities	Payload is a structured object; HAS_PART edges allowed	ARTIFACT, CONTEXT, POLICY, ROLE, CAPABILITY
Relational	Entities whose meaning is primarily relational	Semantic weight defined by relationship edges, not payload alone	CONCEPT, INTENT, CLAIM, CONSTRAINT

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Primitive Class	Description	Key Behavioral Contract	Derived Entity Types
Temporal	Time-bound states or occurrences	Mandatory temporal scope; validity window enforced at runtime	EVENT, STATE

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3. Semantic Graph Model

3.1 Graph Definition

A URIB Semantic Graph is formally defined as the 4-tuple:

$$G = (V, E, \Sigma, \Phi)$$

where:

V

= finite set of entity nodes, each uniquely identified by a NodeID (UUID v4)

E

$$\subseteq V \times V \times R$$

= set of typed directed edges, where R is the set of permitted RelationshipTypes

Σ

= type schema: a mapping from SemanticType identifiers to payload schemas and allowed relationship sets

Φ

= semantic annotation function: $V \cup E \rightarrow \text{MetadataRecord}$, encoding provenance, confidence, temporal scope, and commitment contracts

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A URIB runtime MUST maintain G as a consistent structure satisfying all invariants defined in Section 6 at all times. Mutations to G that would violate any invariant MUST be rejected before application.

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3.2 Node Anatomy

Every node $v \in V$ MUST carry the following mandatory fields:

Field	Type	Required	Description
NodeID	UUID v4	MUST	Globally unique identifier. Stable for the lifetime of the graph version.
SemanticType	Enum (Section 4)	MUST	The node's type from the normative Entity Taxonomy. Determines valid payload schema and allowed edges.
Payload	Typed object	MUST	The typed data content of the node. MUST conform to the schema declared by SemanticType in Σ .
Provenance	ProvenanceRecord	MUST	Origin context: source system, source timestamp, ingestion method, originating agent NodeID.
Confidence	Float [0.0–1.0]	MUST	Semantic certainty score. 1.0 = fully authoritative. Decays through derivation chains per MPR-005.
CreatedAt	ISO 8601 UTC timestamp	MUST	Creation timestamp. Immutable after node creation.
MutabilityClass	Enum	MUST	One of: <code>Immutable</code> <code>Mutable</code> <code>Ephemeral</code> . Governs update and purge behavior.
TemporalScope	TimeRange (optional)	SHOULD	Validity window: <code>valid_from</code> / <code>valid_to</code> in ISO 8601 UTC. Required for Temporal primitive types.
Contracts	List<ContractRef>	MAY	References to active Semantic Commitment Contracts (Section 8) governing this node.
SemanticDomain	DomainID	MUST	The Semantic Domain subgraph this node belongs to. Required for cross-domain bridge validation.

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3.3 Edge Anatomy

Every edge $e \in E$ MUST carry the following fields:

Field	Type	Required	Description
EdgeID	UUID v4	MUST	Globally unique edge identifier.
SourceNodeID	UUID v4	MUST	NodeID of the source (tail) node. MUST exist in V or a declared external boundary.
TargetNodeID	UUID v4	MUST	NodeID of the target (head) node. MUST exist in V or a declared external boundary.
RelationshipType	Enum (Section 5)	MUST	The edge's type from the normative Relationship Taxonomy. MUST be valid for the source and target SemanticTypes.
Directionality	Enum	MUST	One of: <code>Unidirectional</code> <code>Bidirectional</code> <code>Asymmetric</code> .
Weight	Float [0.0–1.0]	MUST	Semantic strength of the relationship. 1.0 = definitive. For Bidirectional edges, MUST be symmetric (MPR-007).
TemporalScope	TimeRange (optional)	SHOULD	Validity window of this relationship. Edge is semantically inactive outside this window.
MutabilityClass	Enum	MUST	One of: <code>Immutable</code> <code>Mutable</code> <code>Ephemeral</code> .
DomainBridge	DomainBridgeRecord (optional)	MUST if cross-domain	Required for any edge crossing a Semantic Domain boundary. See INV-009 and Section 9.2.

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3.4 Graph Composition Rules

When two subgraphs G_1 and G_2 are merged into a composite graph G_3 , the following rules apply:

- 4. **Node identity union:** If a node exists in both G_1 and G_2 with the same NodeID, they **MUST** be treated as the same node. Payload conflicts are resolved by the active merge policy (see below).
- 5. **Edge union:** Edges from both graphs are included. Duplicate EdgeIDs **MUST** be deduplicated; conflicting edge attributes are resolved by merge policy.
- 6. **Invariant re-validation:** After merge, all invariants **MUST** be re-validated on the composite graph.
The merge operation **MUST** be atomic — it either completes fully or is entirely rejected.

Conflict Resolution Policies:

Policy	Rule	When to Use
Newest-Wins	Node with the later CreatedAt timestamp is authoritative	High-velocity event streams; ephemeral state updates
Confidence-Weighted	Node with the higher Confidence value is authoritative	ML-derived entities; claim resolution (default for CLAIM nodes)
Explicit Merge Policy	A POLICY node governing the domain defines the merge procedure	Regulated domains; audit-critical data; cross-organizational merges
Authority-Wins	Node originating from a designated Semantic Authority AGENT wins	Master data management; reference data governance

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3.5 Subgraph Boundaries — Semantic Domains

A URIB Semantic Graph is partitioned into named subgraphs called **Semantic Domains**. A Semantic Domain is a namespaced subgraph with its own type schema version (Σ_d), its own set of governing POLICY nodes, and its own Semantic Authority. Every node MUST belong to exactly one Semantic Domain.

Crossing a domain boundary — creating an edge where `SourceNodeID.SemanticDomain` $\neq$ `TargetNodeID.SemanticDomain` — MUST be accompanied by a **Bridge Operation**. The Bridge Operation creates a `DomainBridgeRecord` specifying:

- Source domain identifier and schema version
- Target domain identifier and schema version
- Type mapping declarations for any types that differ across domains
- The active Meaning-Preservation Contract (minimally SC-CROSS-DOMAIN-FIDELITY)
- A semantic equivalence check result (pass / fail / conditional)

Without a valid `DomainBridgeRecord`, the cross-domain edge MUST be rejected (INV-009).

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4. Entity Type Taxonomy

The following table constitutes the normative Entity Type Registry for URIB v1.0. All implementations MUST support these types. Implementations MAY define extended types within their Semantic Domain schema, but extended types MUST declare their root primitive class and MUST NOT contradict invariants applicable to the base type.

Type Name	Primitive Class	Description	Allowed Payload Structure	Default Mutability	Example Instance
CONCEPT	Relational	An abstract idea, domain concept, or classification category. Meaning is constituted primarily by IS_A and HAS_PART relationships.	label: string, definition: string, synonyms: string[]	Immutable	"Customer Loyalty", "Risk Threshold"
ARTIFACT	Composite	A concrete created object: document, file, database record, message, contract, report. Has a lifecycle and an authoring AGENT.	name: string, mime_type: string, content_ref: URI, version: string	Mutable	Invoice PDF, API response record, configuration file
AGENT	Composite	An actor with declared intent and the capacity for action. May be a human user, an automated service, an AI system, or an organizational unit.	agent_id: string, agent_type: enum(human system org), display_name: string	Mutable	User "usr-8821", Service "payment-processor-v2"
EVENT	Temporal	A discrete occurrence at a point or short interval in time. Atomic and non-repeating. Has a definite temporal anchor.	event_type: string, occurred_at: ISO8601, source_system: string	Immutable	Purchase transaction, user login, sensor trigger, deployment

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Type Name	Primitive Class	Description	Allowed Payload Structure	Default Mutability	Example Instance
STATE	Temporal	A condition that persists over a duration. Has a start and (optionally) an end. Represents the world between events.	state_type: string, valid_from: ISO8601, valid_to: ISO8601 null	Mutable	"Account suspended", "Order in transit", "Service degraded"
METRIC	Atomic	A measured quantity with an explicit unit, measurement timestamp, and temporal scope. MUST specify what entity it MEASURES.	value: number, unit: string, measured_at: ISO8601, precision: number	Immutable	Revenue: 149.99 USD, Latency: 43ms, Score: 0.87
INTENT	Relational	A declared purposeful direction of action, attributed to an AGENT. Distinguishes deliberate action from automated system behavior.	intent_label: string, goal_description: string, declared_at: ISO8601	Immutable	"Purchase intent", "Escalation intent", "Optimization goal"
CONSTRAINT	Relational	A limiting rule, boundary condition, or compliance requirement that CONSTRAINS the behavior of other entities.	constraint_type: string, rule_expression: string, severity: enum	Mutable	"Max transaction: \$10,000", "Response time ≤ 200ms", "GDPR retention limit"
REFERENCE	Atomic	A pointer to another node within the graph or to an external resource outside the graph boundary. Carries no semantic payload beyond the pointer itself.	ref_target: UUID URI, ref_type: enum(internal external), resolved: bool	Mutable	External URL, cross-domain node pointer, foreign key mapping
SIGNAL	Atomic	A transient data pulse with a semantic label. Typically ephemeral. May be elevated to a	signal_label: string, raw_value: any, emitted_at: ISO8601, ttl_seconds: integer	Ephemeral	Sensor reading, webhook ping, alert trigger,

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Type Name	Primitive Class	Description	Allowed Payload Structure	Default Mutability	Example Instance
		STATE node when threshold conditions are met (MPR-009).			health check result
CLAIM	Relational	An asserted proposition with declared confidence and provenance. May be contested by another CLAIM via CONTRADICTS. Subject to conflict resolution (INV-011).	claim_text: string, confidence: float, claimed_by: AgentID, basis: string	Mutable	"Customer is high-value" (conf: 0.91), "Model prediction: churn" (conf: 0.74)
CONTEXT	Composite	A semantic scope that frames the interpretation of other entities. Child entities within a CONTEXT inherit its interpretive frame (MPR-006).	context_name: string, frame_type: string, inherited_from: ContextID null	Mutable	Fiscal year 2026 context, EU regulatory context, A/B test cohort context
ROLE	Composite	A relational position that an AGENT occupies within a CONTEXT. Separates the AGENT's identity from the function it performs in a given setting.	role_name: string, role_type: string, context_id: ContextID	Mutable	"Account Owner", "Approver", "Observer", "Orchestrator"
CAPABILITY	Composite	A declared ability belonging to an AGENT or ARTIFACT. Represents what the entity can do, not what it is doing. May be GOVERNED by a POLICY.	capability_name: string, capability_type: string, scope: string, version: string	Mutable	"Can process refunds", "Supports OAuth 2.0", "ML inference: classification"
POLICY	Composite	A rule set governing operations within a Semantic Domain. A POLICY GOVERNS entities and is evaluated before any	policy_id: string, policy_type: string, rules: Rule[], version: string, effective_from: ISO8601	Immutable	Data retention policy, access control policy, semantic merge policy

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Type Name	Primitive Class	Description	Allowed Payload Structure	Default Mutability	Example Instance
		mutation to those entities (MPR-008).			

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5. Relationship Type Taxonomy

The following table constitutes the normative Relationship Type Registry for URIB v1.0. Every edge in the semantic graph **MUST** use one of these RelationshipTypes. The "Valid Source Types" and "Valid Target Types" columns define the permitted type combinations — using any combination not listed is a Type Violation (INV-004).

Relationship	Direction	Semantics	Valid Source Types	Valid Target Types	Invariant Notes
IS_A	Unidirectional	Subtype or classification. Asserts that source is a member of the target category.	CONCEPT, AGENT, ARTIFACT, ROLE	CONCEPT	MUST NOT create cycles in IS_A subgraph (DAG required)
HAS_PART	Unidirectional	Compositional containment. Source contains target as a structural component.	ARTIFACT, CONTEXT, POLICY, EVENT	Any	No nested cycles; MUST NOT imply ownership transfer
CAUSES	Unidirectional	Causal dependency. Source event/state directly produces target event/state.	EVENT, STATE	EVENT, STATE	INV-005: CAUSES subgraph MUST be a DAG; no causal cycles
PRECEDES	Unidirectional	Strict temporal ordering. Source occurs before target with no temporal overlap.	EVENT, STATE	EVENT, STATE	INV-006: Source temporal scope MUST end before target begins
REFERENCES	Unidirectional	Non-owning pointer relationship.	Any	Any, REFERENCE	REFERENCE nodes MUST resolve; dangling refs raise INV-003

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Relationship	Direction	Semantics	Valid Source Types	Valid Target Types	Invariant Notes
		Source refers to target without structural dependency.			
OWNS	Unidirectional	Ownership or possession. Source AGENT is the declared owner of target.	AGENT	ARTIFACT, CAPABILITY, ROLE	An entity MUST NOT have more than one OWNS-parent unless multi-ownership is declared
PERFORMS	Unidirectional	Agent action execution. Source AGENT is the actor that executed the target EVENT.	AGENT	EVENT	PERFORMS MUST be accompanied by provenance timestamp in the edge record
INHABITS	Unidirectional	Agent occupying a role within a context. Separates agent identity from functional position.	AGENT	ROLE	ROLE MUST be SCOPED by a CONTEXT; INHABITS without CONTEXT is a modeling warning
GOVERNS	Unidirectional	Policy authority. Source POLICY applies governance rules to target entity. Evaluated before every mutation to the target.	POLICY	Any	MPR-008: GOVERNS evaluation is mandatory before write; cannot be bypassed
SATISFIES	Unidirectional	Fulfillment of a constraint. Source entity demonstrates compliance with the target CONSTRAINT.	Any	CONSTRAINT	SATISFIES claims MUST carry confidence score; unvalidated SATISFIES has weight ≤ 0.5
CONTRADICTS	Bidirectional	Logical conflict between two CLAIMs. Both cannot be simultaneously	CLAIM	CLAIM	INV-011: Resolution procedure MUST be invoked; symmetric weight required (MPR-007)

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Relationship	Direction	Semantics	Valid Source Types	Valid Target Types	Invariant Notes
		canonical. Triggers INV-011 conflict resolution.			
MEASURES	Unidirectional	Quantification relationship. Source METRIC is a measured characterization of the target entity.	METRIC	Any (except SIGNAL, METRIC)	MUST include unit and measurement timestamp in METRIC payload
DERIVES_FROM	Unidirectional	Provenance chain. Source was produced by transformation or derivation from target. Confidence MUST decay (MPR-005).	Any	Any	INV-010: Chain MUST be traceable to authoritative root; MPR-005 applies at every hop
TRANSFORMS_TO	Unidirectional	Output of a transformation operation. Source is the input; target is the result. TRANSFORMS_TO implies a declared transformation procedure.	Any	Any	Transformation procedure MUST be declared in edge annotations; MPR-001 applies
SCOPES	Unidirectional	Context scoping. Source CONTEXT provides the interpretive frame for target entity. MPR-006 context inheritance applies.	CONTEXT	Any	CONTEXT scope MUST NOT conflict with governing POLICY in the same domain
EMITS	Unidirectional	Signal emission. Source AGENT or ARTIFACT is the emitter of the target SIGNAL. Records the signal	AGENT, ARTIFACT	SIGNAL	INV-012: SIGNAL TTL MUST be set; ephemeral purge enforced after TTL expiry

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Relationship	Direction	Semantics	Valid Source Types	Valid Target Types	Invariant Notes
		origin for provenance tracking.			
INTENDS	Unidirectional	Intent declaration. Source AGENT has declared the target INTENT. Propagated to ARTIFACT nodes per MPR-004.	AGENT	INTENT	MPR-004: INTENT propagates to ARTIFACT nodes within the same CONTEXT scope
CONSTRAINS	Unidirectional	Active constraint application. Source CONSTRAINT applies its limiting rule to target entity. Evaluated at mutation time.	CONSTRAINT	Any	Violations of a CONSTRAINS relationship MUST be raised as ConstraintViolationErrors

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6. Semantic Invariants

Semantic Invariants are runtime-enforced constraints that **MUST** hold for all states of the semantic graph G . A URIB runtime **MUST** evaluate all applicable invariants before committing any graph mutation. If any invariant would be violated, the mutation **MUST** be rejected atomically before any changes are applied to the graph.

Invariants are categorized by violation consequence: **REJECT** (operation fails, graph unchanged), **WARN** (operation proceeds with warning), and **DEMOTE** (node status reduced). **MUST**-level invariants produce **REJECT** outcomes.

INV-001 Node Identity Uniqueness

Statement: Every node $v \in V$ **MUST** have a globally unique NodeID within the graph. No two nodes in the same graph **MUST** share a NodeID value.

Formal: $\forall v_1, v_2 \in V : v_1.\text{NodeID} = v_2.\text{NodeID} \implies v_1 = v_2$

Rationale: NodeID is the universal identity anchor. Duplicate NodeIDs create ambiguous reference resolution, break provenance chains, and corrupt edge endpoint lookups.

Violation Consequence: **REJECT** — duplicate node **MUST** be rejected. Conflict logged with both NodeIDs and the rejecting domain.

INV-002 Type Conformance

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Statement: Every node's Payload MUST conform to the schema declared by its SemanticType in the type schema Σ . All required payload fields MUST be present and correctly typed.

Rationale: Type conformance is the foundation of semantic interoperability. A METRIC node with a missing unit field cannot be safely interpreted by any consuming system.

Violation Consequence: REJECT — deserialization or schema validation failure. Error MUST include the failing field path and the expected schema.

INV-003 Edge Endpoint Existence

Statement: Every edge $e \in E$ MUST reference a SourceNodeID and TargetNodeID that exist in V or in a declared external reference boundary (a registered ExternalDomain record). Dangling edges — edges referencing non-existent nodes — MUST NOT be persisted.

Rationale: Dangling edges break graph traversal, provenance reconstruction, and policy evaluation. An edge to a non-existent node cannot carry semantic meaning.

Violation Consequence: REJECT — edge is dropped. An IntegrityError MUST be raised identifying the missing NodeID and the edge that references it.

INV-004 Relationship Validity

Statement: Every edge's RelationshipType MUST be permitted between the SemanticTypes of its SourceNode and TargetNode, as defined in the Relationship Taxonomy (Section 5). Type combinations not listed in the taxonomy are forbidden.

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Rationale: Invalid relationship types produce semantically nonsensical graphs. An AGENT CAUSES POLICY has no defined meaning in the URIB model and cannot be evaluated by consuming systems.

Violation Consequence: REJECT — `TypeViolationError` raised. Error MUST specify the source type, target type, relationship type, and the taxonomy entry that was violated.

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INV-005 Acyclicity in Causal Chains

Statement: The subgraph $G_{\text{causes}} = (V, E_{\text{causes}})$ where $E_{\text{causes}} = \{e \in E : e.\text{RelationshipType} = \text{CAUSES}\}$ MUST form a Directed Acyclic Graph (DAG). Cycles in the causal subgraph indicate a fundamental modeling error.

Rationale: Causal cycles are logically incoherent (Event A causes Event B causes Event A). They also produce infinite loops in causal chain traversal algorithms.

Violation Consequence: REJECT — `CyclicCausalChainError` raised. The cycle MUST be identified and reported (list of NodeIDs forming the cycle). The graph mutation that would create the cycle is rejected.

INV-006 Temporal Consistency

Statement: For any edge e with $\text{RelationshipType} = \text{PRECEDES}$ from node A to node B: $A.\text{TemporalScope}.\text{valid_to}$ MUST be strictly less than $B.\text{TemporalScope}.\text{valid_from}$. No temporal overlap is permitted. If either node lacks a declared `TemporalScope`, the `PRECEDES` edge is a modeling warning.

Rationale: A `PRECEDES` relationship makes a temporal ordering assertion. Without temporal consistency enforcement, ordering claims become unverifiable.

Violation Consequence: REJECT — `TemporalContradictionError` raised for overlapping scopes. WARN raised if `TemporalScope` is absent on either node.

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INV-007 Claim Confidence Bounds

Statement: Every node with `SemanticType = CLAIM` MUST have a `Confidence` value in the closed interval `[0.0, 1.0]`. Values outside this interval are invalid.

Rationale: Confidence values outside `[0.0, 1.0]` cannot be interpreted as probabilities or relative certainty scores. They break all downstream reasoning that consumes confidence values.

Violation Consequence: WARN — value is clamped to `[0.0, 1.0]` with a normalization warning logged. The originating system MUST be notified of the normalization.

INV-008 Immutable Node Integrity

Statement: Nodes with `MutabilityClass = Immutable` MUST NOT be modified after their initial creation and persistence. Any attempted mutation — payload change, type change, metadata update — MUST be rejected.

Rationale: Immutable nodes serve as the stable anchor points of the semantic graph. Mutation of immutable nodes breaks provenance chains, invalidates historical audit records, and undermines Semantic Commitment Contracts.

Violation Consequence: REJECT — `MutationForbiddenError` raised. Error MUST include the `NodeID`, the attempted mutation, and the identity of the requesting AGENT.

INV-009 Domain Bridge Declaration

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Statement: Any edge where `SourceNode.SemanticDomain` $\neq$ `TargetNode.SemanticDomain` MUST carry a valid `DomainBridgeRecord` in its edge annotations. The `DomainBridgeRecord` MUST specify source domain, target domain, schema version mapping, and the active `MeaningPreservationContract` reference.

Rationale: Cross-domain edges without bridge declarations silently carry meaning across semantic boundaries that may be incompatible. The `DomainBridge` declaration makes this crossing explicit and machine-verifiable.

Violation Consequence: REJECT — `CrossDomainEdgeRejectionError` raised. The edge is not persisted. The requesting system MUST perform a Bridge Operation before re-submitting.

INV-010 Provenance Chain Completeness

Statement: Any `DERIVES_FROM` chain MUST be traceable, by following `DERIVES_FROM` edges backward, to a root node whose `Provenance.origin` is either "authoritative-source" or "user-input". Orphaned provenance chains — chains that terminate at a node with unknown or missing origin — MUST NOT be promoted to authoritative status.

Rationale: A claim or artifact whose provenance cannot be traced has unknown reliability. Allowing unverifiable provenance to attain authoritative status undermines the entire meaning-preservation model.

Violation Consequence: WARN + DEMOTE — node is tagged as "unverified-provenance" and its Confidence is capped at 0.5 until provenance is established.

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INV-011 Contradiction Isolation

Statement: Two CLAIM nodes connected by a CONTRADICTS edge MUST NOT both carry canonical status simultaneously. When a CONTRADICTS edge is created, the graph MUST immediately invoke the active conflict resolution procedure for the governing domain. Until resolution completes, neither CLAIM may be used in downstream reasoning as canonical.

Rationale: Allowing contradictory canonical claims produces non-deterministic behavior in any system that consumes the graph. Contradiction must be an exceptional, immediately-resolved state.

Violation Consequence: HALT — ContradictionHaltError raised. Resolution procedure invoked. Both CLAIMs suspended from canonical use pending resolution outcome.

INV-012 Signal Ephemerality

Statement: All SIGNAL nodes with MutabilityClass = Ephemeral MUST be purged from the live semantic graph after their declared TTL (time-to-live) in seconds expires. SIGNAL nodes without a declared TTL MUST default to a runtime-configured maximum ephemeral TTL (default: 300 seconds).

Rationale: Stale SIGNAL nodes represent outdated transient data. Allowing them to persist in the live graph creates false positives in signal-driven workflows and pollutes the live truth surface with historical noise.

Violation Consequence: INVALIDATE — stale SIGNAL node is marked as expired and removed from the live graph. Consuming edges from expired SIGNALs are severed and logged as StaleSignalErrors.

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7. Meaning-Preservation Rules

Meaning-Preservation Rules (MPRs) govern how semantic fidelity is maintained across graph operations: transformations, derivations, serializations, and cross-domain transfers. Unlike Invariants (which are binary pass/fail gates), MPRs define behavioral expectations for how operations **MUST** be conducted. Violations of **MUST**-level MPRs **MUST** be rejected; violations of **SHOULD**-level MPRs produce warnings.

MPR-001 Transformation Transparency

Rule: Any graph operation that produces a new node from one or more existing nodes **MUST** record the lineage relationship via a `TRANSFORMS_TO` or `DERIVES_FROM` edge. The transformation procedure (ETL step, ML inference, aggregation function, schema mapping) **MUST** be declared in the edge annotations.

Rationale: Without transformation lineage, the semantic graph becomes a collection of unconnected facts with no traceable history. Meaning cannot be preserved if its derivation path is unknown.

Level: **MUST**. Violation causes **REJECT** of the new node creation.

MPR-002 Type Promotion Constraints

Rule: A node's `SemanticType` **MAY** be promoted to a more specific type (e.g., `ARTIFACT` → a domain-specific subtype of `ARTIFACT`) if the payload schema of the new type is a strict superset of the current schema. A node's `SemanticType` **MUST NOT** be demoted to a less specific type unless

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explicitly authorized by a Semantic Authority AGENT with a GOVERNS relationship over the node's domain.

Rationale: Type demotion silently discards semantic precision. An EVENT node being reclassified as a SIGNAL loses its temporal ordering guarantees and immutability contract.

Level: MUST NOT for demotion without authority. Unauthorized demotion MUST be rejected with TypeDemotionError.

MPR-003 Payload Losslessness

Rule: When a node's payload is serialized for cross-domain transfer or storage, all semantically significant fields MUST be preserved. If lossy serialization is technically unavoidable (e.g., precision reduction, encoding limitation), it MUST be explicitly declared in the DomainBridgeRecord as a LossDeclaration with field-level detail.

Rationale: Silent payload loss is the most common form of semantic degradation in integration systems. Making lossiness explicit — even when unavoidable — preserves the ability of consuming systems to reason correctly about what they received.

Level: MUST for lossless preservation; MUST declare loss if unavoidable. Silent loss MUST be treated as a violation.

MPR-004 Intent Propagation

Rule: An INTENT node associated with an AGENT via an INTENDS edge MUST be propagated — via REFERENCES edges — to all ARTIFACT nodes produced by that AGENT via PERFORMS within the

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same CONTEXT scope, unless an explicit intent suppression annotation is present on the PERFORMS edge.

Rationale: The intent behind an action is part of the meaning of the artifacts it produces. An invoice generated with the intent to defraud has different semantic status than one generated with a payment-processing intent, even if the byte content is identical.

Level: MUST unless suppressed. Suppression MUST be logged with the suppressing AGENT's NodeID.

MPR-005 Confidence Decay on Derivation

Rule: When a new CLAIM or METRIC node is derived from a parent node via a DERIVES_FROM edge, the child's Confidence MUST NOT exceed the parent's Confidence multiplied by a declared decay factor δ (default $\delta = 0.95$). The decay factor MUST be declared in the DERIVES_FROM edge annotations. Unjustified confidence amplification — where a child's confidence exceeds the parent's — is prohibited.

Rationale: Every derivation step introduces potential error. A chain of derived claims that does not decay in confidence creates an epistemically unjustifiable certainty amplification.

Level: MUST. Child Confidence = $\min(\text{declared_confidence}, \text{parent_confidence} \times \delta)$. Amplification MUST be rejected.

MPR-006 Context Inheritance

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Rule: Child nodes within a CONTEXT scope (connected via SCOPES edges) inherit the interpretive frame of that CONTEXT unless overridden by a more specific locally-scoped CONTEXT. When a child node overrides an inherited CONTEXT, the override MUST be logged with the overriding CONTEXT's NodeID, the original CONTEXT's NodeID, and the requesting AGENT.

Rationale: Context inheritance ensures that nodes don't need to redundantly declare their interpretive frame. The CONTEXT-as-scope model is only useful if inheritance is reliably applied.

Level: MUST for inheritance; MUST log override. Silent override is a semantic integrity warning.

MPR-007 Relationship Semantic Symmetry

Rule: For any edge with Directionality = Bidirectional, the semantic Weight MUST be equal in both traversal directions. Asymmetric weight assignment on a Bidirectional edge (e.g., Weight = 0.9 forward, Weight = 0.6 reverse) is a structural modeling error and MUST be detected and rejected at edge creation time.

Rationale: Bidirectionality asserts semantic equivalence in both directions. Asymmetric weights on a bidirectional edge produce inconsistent graph traversal results depending on traversal direction.

Level: MUST. AsymmetricBidirectionalEdgeError raised if Weight is not symmetric.

MPR-008 Policy Supremacy

Rule: Any POLICY node connected to an entity via a GOVERNS edge MUST be evaluated before any mutation to that entity (including payload updates, edge additions, type changes, and deletion).

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Mutations that would violate an active POLICY MUST be rejected before execution. There is no mechanism to bypass policy evaluation.

Rationale: POLICY nodes represent governance commitments: compliance rules, data sovereignty requirements, access controls. A semantic model where policies can be bypassed provides no governance guarantees.

Level: MUST. PolicyViolationError raised. The violating mutation is rejected and the policy rule that was violated MUST be identified in the error.

MPR-009 Signal-to-State Elevation

Rule: A SIGNAL node MAY be elevated to a STATE node only when two conditions are simultaneously met: (1) the SIGNAL's value satisfies a declared threshold rule registered in the governing domain, and (2) the SIGNAL's Confidence meets or exceeds the domain's elevation confidence floor (default: 0.80). Premature elevation — before both conditions are verified — is prohibited.

Rationale: SIGNALs are transient; STATEs are persistent and carry greater semantic weight (they affect ordering, constraint evaluation, and policy application). Unvalidated elevation conflates transient observations with authoritative conditions.

Level: MUST satisfy both conditions. PrematureElevationError raised if elevation is attempted before conditions are met.

MPR-010 Semantic Version Locking

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Rule: When a Semantic Domain's type schema Σ is updated to a new version, all existing nodes conforming to the prior schema version **MUST** be tagged with their schema version identifier (e.g., `schema_version: "v1.0"`). Cross-version graph operations — operations that involve nodes from different schema versions — **MUST** include a schema compatibility check. Operations that fail compatibility checks **MUST** be rejected or routed through a schema migration procedure.

Rationale: Schema evolution without version tracking produces silent semantic drift — the same field name means different things in different schema versions, producing incorrect downstream interpretations.

Level: **MUST** for version tagging. **MUST** check compatibility on cross-version operations. `SchemaVersionConflictError` raised on incompatible cross-version access.

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8. Semantic Commitment Contracts

8.1 Definition

A **Semantic Commitment Contract** (SCC) is a formal, machine-evaluable declaration made at graph construction time — or at the time of a node's elevation to a higher-integrity status — that a node or edge will maintain specified semantic properties across its lifetime. Contracts are not documentation: they are runtime promises that **MUST** be enforced by the URIB runtime at every relevant operation.

A node **MAY** carry multiple contracts simultaneously. When contracts conflict, the more restrictive contract takes precedence. Contracts are stored in the node's `Contracts` field as `ContractRef` records, each containing: `ContractType`, `IssuedAt`, `IssuedBy` (AGENT NodeID), and `ContractParameters` (contract-specific configuration).

SC-IDENTITY

Declaration: "This node represents exactly one real-world entity for the lifetime of this graph version. Its semantic identity is stable and singular."

Commits:

- NodeID stability: NodeID **MUST NOT** change for any reason
- Type immutability: SemanticType **MUST NOT** be changed after contract activation
- No merge: This node **MUST NOT** be merged with another node without explicit contract invalidation

Enforcement Point: Every node mutation. Every merge operation involving this node.

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SC-MEANING-STABLE

Declaration: "The semantic interpretation of this node's payload will not change as the domain's type schema evolves. Consumers may rely on this node's meaning remaining consistent across schema versions."

Commits:

- Backward-compatible payload structure: new schema versions MUST NOT remove or reinterpret this node's existing payload fields
- No silent re-semanticization: if a field's meaning changes in a new schema version, this node MUST be explicitly migrated or its contract invalidated

Enforcement Point: Every schema version upgrade in the governing domain. MPR-010 applies.

SC-PROVENANCE-AUDITABLE

Declaration: "The complete derivation chain of this node can be reconstructed at any time from the live graph's DERIVES_FROM edges and provenance records."

Commits:

- DERIVES_FROM chain completeness: no link in the provenance chain may be deleted while this contract is active
- Retention of provenance records: Provenance.ProvenanceRecord MUST be retained for the contract lifetime

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- Auditability on demand: the full chain **MUST** be reconstructable via a standard graph traversal query

Enforcement Point: Any deletion of a DERIVES_FROM edge or a provenance-chain ancestor node is blocked if it would break this contract.

SC-CROSS-DOMAIN-FIDELITY

Declaration: "When this node crosses a Semantic Domain boundary, its meaning will be fully preserved by the receiving domain. Any loss or approximation **MUST** be declared."

Commits:

- Lossless serialization by default (MPR-003 applies)
- Type mapping declaration in the DomainBridgeRecord
- Semantic equivalence check **MUST** be performed and recorded
- If the receiving domain cannot achieve full fidelity, a LossDeclaration **MUST** be attached and the node's Confidence is adjusted accordingly

Enforcement Point: Every Bridge Operation involving this node. INV-009 is a prerequisite gate.

SC-TEMPORAL-BOUNDED

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Declaration: "This node is semantically valid only within its declared TemporalScope. Outside that scope, it MUST NOT be interpreted as authoritative and MUST NOT be included in current-state graph views."

Commits:

- TTL enforcement: the runtime MUST purge or archive this node at TemporalScope.valid_to
- Expiry notification: consuming systems MUST be notified of expiry before the expiry time
- No retroactive extension: valid_to MUST NOT be extended without re-issuance of the contract by a Semantic Authority

Enforcement Point: Every current-state query, every policy evaluation, every CLAIM that derives from this node. INV-012 applies for SIGNAL nodes.

SC-POLICY-GOVERNED

Declaration: "All mutations to this node are subject to the referenced POLICY node. No write operation may bypass policy evaluation."

Commits:

- Policy evaluation before every write (MPR-008 applies)
- POLICY node MUST be active and resolvable at evaluation time
- If the governing POLICY node is deleted or invalidated, all SC-POLICY-GOVERNED contracts referencing it MUST raise a PolicyResolutionError

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Enforcement Point: Every payload mutation, edge addition/removal, type change, and deletion attempt on this node.

SC-CONFIDENCE-FLOOR

Declaration: "This node will never participate in downstream reasoning as a canonical input if its Confidence drops below the declared floor value."

Commits:

- ContractParameters MUST include floor_value: float in [0.0, 1.0]
- When $\text{Confidence} \leq \text{floor_value}$, node is automatically tagged "below-floor" and excluded from canonical queries
- Below-floor events MUST be logged and the issuing AGENT notified

Enforcement Point: Every Confidence update (including MPR-005 decay propagation). Every canonical query that would include this node.

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SC-CONTRADICTION-FREE

Declaration: "No CONTRADICTS edge targeting this node will be permitted while this contract is active. This node's claim is asserted as contradiction-free within its governing domain."

Commits:

- Any attempt to create a CONTRADICTS edge targeting this node MUST be blocked
- If a competing CLAIM is identified that would normally trigger a CONTRADICTS edge, the contradiction detection MUST be raised as a blocking error rather than silently creating the edge
- This contract MUST be explicitly invalidated by a Semantic Authority before any CONTRADICTS edge may be created

Enforcement Point: Every CONTRADICTS edge creation attempt. INV-011 resolution gate is deferred to contract invalidation time.

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9. Implementation Guide & Examples

9.1 Building a Semantic Token — Step-by-Step

This example walks through the complete semantic tokenization of a raw Commerce API payload. The raw payload contains no type contracts, no provenance, no relationships, and no confidence signals. After tokenization, it becomes a typed subgraph of four interconnected semantic tokens with full invariant coverage and committed contracts.

Input: Raw API Payload

```
{  "user_id":    "usr-8821",  "action":      "purchase",  "item_sku":    "SKU-4490",  "timestamp":   "2026-07-09T00:00:00Z",  "amount":      149.99 }
// No type context. No provenance. No relationships. No confidence. // No mutability contract. No domain. No invariants applied.
```

Output: Semantic Token Subgraph

Node 1 — AGENT (the customer):

```
{  "NodeID":      "a1b2c3d4-0001-4000-8000-000000000001",
  "SemanticType": "AGENT",  "Payload": {    "agent_id":      "usr-8821",
  "agent_type":    "human",  "display_name":    "Customer usr-8821"  },
  "Provenance": {    "origin":      "authoritative-source",
  "source_system": "commerce-api-v3",  "ingested_at":    "2026-07-09T00:00:01Z"  },
  "Confidence":    1.0,  // authoritative-source origin
  "CreatedAt":     "2026-07-09T00:00:01Z",  "MutabilityClass": "Mutable",
  "SemanticDomain": "commerce-domain-v1",  "Contracts":      ["SC-IDENTITY"] }
```

Node 2 — EVENT (the purchase event):

```
{  "NodeID":      "a1b2c3d4-0002-4000-8000-000000000002",
  "SemanticType": "EVENT",  "Payload": {    "event_type":    "purchase",
  "occurred_at":   "2026-07-09T00:00:00Z",  "source_system": "commerce-api-v3"  },
  "TemporalScope": {    "valid_from":    "2026-07-09T00:00:00Z",
  "valid_to":      "2026-07-09T00:00:00Z"  // point event
  },
  "Confidence":    1.0,  "MutabilityClass": "Immutable",  // EVENT nodes are Immutable (INV-008)
  "SemanticDomain": "commerce-domain-v1",
  "Contracts":     ["SC-TEMPORAL-BOUNDED", "SC-PROVENANCE-AUDITABLE"] }
```

Node 3 — ARTIFACT (the purchased item):

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```
{  "NodeID":          "a1b2c3d4-0003-4000-8000-0000000000003",
  "SemanticType":     "ARTIFACT",    "Payload": {      "name":          "SKU-
4490",      "mime_type":        "application/x-product-sku",      "content_ref":
"sku://catalog/SKU-4490",      "version":          "current"    },
  "MutabilityClass":  "Immutable",   "SemanticDomain":  "commerce-domain-v1",
  "Contracts":        ["SC-IDENTITY"] }
```

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Node 4 — METRIC (the transaction amount):

```
{  "NodeID":          "a1b2c3d4-0004-4000-8000-000000000004",
  "SemanticType":     "METRIC",   "Payload": {      "value":          149.99,
  "unit":             "USD",      "measured_at":      "2026-07-09T00:00:00Z",
  "precision":        2 // decimal places },  "MutabilityClass":
  "Immutable",  "SemanticDomain": "commerce-domain-v1" }
```

Edges (the semantic relationships):

```
// Edge 1: AGENT PERFORMS EVENT { "RelationshipType": "PERFORMS",
"SourceNodeID": "...0001", "TargetNodeID": "...0002",  "Weight": 1.0,
"Directionality": "Unidirectional" } // Edge 2: EVENT HAS_PART ARTIFACT {
"RelationshipType": "HAS_PART",  "SourceNodeID": "...0002", "TargetNodeID":
"...0003",  "Weight": 1.0, "Directionality": "Unidirectional" } // Edge 3:
METRIC MEASURES EVENT { "RelationshipType": "MEASURES",  "SourceNodeID":
"...0004", "TargetNodeID": "...0002",  "Weight": 1.0, "Directionality":
"Unidirectional" }
```

9.2 Cross-Domain Bridge Operation Example

The EVENT node from Section 9.1 (in `commerce-domain-v1`) is now required by the `analytics-domain-v2`. A Bridge Operation is mandatory.

```
// Step 1: Create the DomainBridgeRecord {  "DomainBridgeID":          "bridge-
2026-0709-001",  "SourceDomain":          "commerce-domain-v1",
"TargetDomain":          "analytics-domain-v2",  "SourceSchemaVersion":
"v1.0",  "TargetSchemaVersion": "v2.0",  "TypeMappings": [      {
"sourceType": "EVENT",  "targetType": "EVENT",  // same type,
different schema version  "fieldMappings": {      "event_type":
"event_category", // renamed field in analytics schema  "occurred_at":
"event_ts" // renamed field in analytics schema  }      } ],
"LossDeclarations": [], // no loss declared — lossless bridge
"SemanticEquivalenceCheck": "PASS",  "ActiveContracts": ["SC-CROSS-DOMAIN-
FIDELITY"],  "BridgedAt": "2026-07-09T00:00:05Z",  "AuthorizedBy":
"semantic-authority-agent-NodeID" } // Step 2: Attach the DomainBridgeRecord
to the cross-domain edge {  "RelationshipType": "REFERENCES",
"SourceNodeID":          "analytics-domain-node-NodeID",  "TargetNodeID":
"...0002", // the EVENT node in commerce-domain  "DomainBridge":
"bridge-2026-0709-001", // INV-009 satisfied  "MutabilityClass":
"Immutable" } // Analytics domain receives the node with field mapping
applied. // Confidence unchanged (lossless bridge). SC-CROSS-DOMAIN-FIDELITY
committed.
```

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9.3 Invariant Violation Handling — INV-005 Causal Cycle

A developer attempts to add a CAUSES edge that would create a cycle: Event A → Event B → Event C → Event A.

```
// Existing graph state: // EventA --CAUSES--> EventB --CAUSES--> EventC //
Attempted mutation - would close the cycle: POST /graph/edges {
  "RelationshipType": "CAUSES",    "SourceNodeID": "EventC-NodeID",
  "TargetNodeID": "EventA-NodeID" } // Runtime response - INV-005 triggered: {
  "error":      "CyclicCausalChainError",    "invariant":  "INV-005",
  "message":    "Adding this CAUSES edge would create a cycle in the causal
subgraph.",    "cycle":      ["EventA-NodeID", "EventB-NodeID", "EventC-
NodeID", "EventA-NodeID"],    "resolution": "Re-model as PRECEDES (temporal
ordering without causal claim), or introduce an intermediary CONTEXT node to
break the causal loop.",    "status":      "REJECTED" } // Graph is unchanged.
Mutation was atomically rejected before application.
```

9.4 Confidence Decay — MPR-005 Applied

A root CLAIM node (Confidence = 0.92) generates a chain of derived CLAIMs. Default decay factor δ = 0.95.

Hop	Node	DERIVES_FROM	Max Allowed Confidence	Calculation
0 (root)	CLAIM-ROOT	— (authoritative-source)	0.9200	Source confidence
1	CLAIM-DERIVED-1	CLAIM-ROOT	0.8740	0.9200×0.95
2	CLAIM-DERIVED-2	CLAIM-DERIVED-1	0.8303	0.8740×0.95
3	CLAIM-DERIVED-3	CLAIM-DERIVED-2	0.7888	0.8303×0.95

Implementation Note

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If CLAIM-DERIVED-3 has an SC-CONFIDENCE-FLOOR contract with floor_value = 0.80, it is automatically tagged "below-floor" at hop 3 (0.7888

<

0.80) and excluded from canonical query results. The runtime MUST emit a BelowFloorEvent notification at this point.

9.5 CLAIM Contradiction Resolution

Two CLAIM nodes are created with conflicting assertions about the same AGENT. A CONTRADICTS edge triggers INV-011.

```
// CLAIM-A: "Customer usr-8821 is high-value" - Confidence: 0.91, CreatedAt: 2026-07-08T10:00Z // CLAIM-B: "Customer usr-8821 is low-value" - Confidence: 0.74, CreatedAt: 2026-07-09T00:00Z // A CONTRADICTS edge is created (INV-011 triggered): ContradictionResolutionProcedure.invoke({ "claimA": { "NodeID": "CLAIM-A", "confidence": 0.91, "createdAt": "2026-07-08T10:00Z" }, "claimB": { "NodeID": "CLAIM-B", "confidence": 0.74, "createdAt": "2026-07-09T00:00Z" }, "policy": "confidence-weighted" // domain's active conflict resolution policy }) // Resolution outcome: { "winner": "CLAIM-A", // higher confidence (0.91 > 0.74) "loser": "CLAIM-B", "action": "CLAIM-B demoted to non-canonical; tagged 'superseded'", "claimA_status": "canonical", "claimB_status": "superseded", "resolution_ts": "2026-07-09T00:00:02Z" } // CLAIM-B is retained in the graph for audit purposes (SC-PROVENANCE-AUDITABLE) // but is excluded from all canonical query results.
```

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10. Conformance Levels

URIB conformance is tiered. Each level is a strict superset of the level below it. Implementations **MUST** declare their conformance level and **MUST** pass all required test assertions for that level. The conformance level is included in the implementation's URIB Conformance Declaration (UCD) document.

Level 1 — Core Conformant

What is required:

- Full implementation of the Entity Type Taxonomy (Section 4): all 15 base types **MUST** be supported
- Full implementation of the Relationship Type Taxonomy (Section 5): all 18 relationship types **MUST** be supported
- All **MUST**-level Semantic Invariants enforced (INV-001 through INV-012)
- Node anatomy with all **MUST** fields populated on every node creation
- Edge endpoint existence validation (INV-003) on every edge creation

Expected Tooling: Graph storage engine with schema validation; invariant checker executable at write time; UUID v4 generator; ISO 8601 timestamp handling.

Required Test Assertions:

- T1-001: Attempt to create a duplicate NodeID → **MUST** be rejected (INV-001)
- T1-002: Attempt to create a node with invalid payload → **MUST** be rejected (INV-002)
- T1-003: Attempt to create a dangling edge → **MUST** be rejected (INV-003)
- T1-004: Attempt an invalid relationship type combination → **MUST** be rejected (INV-004)
- T1-005: Attempt to create a causal cycle → **MUST** be rejected (INV-005)

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- T1-006: Attempt to mutate an Immutable node → MUST be rejected (INV-008)
 - T1-007: All 15 entity types can be created with conformant payloads
 - T1-008: All 18 relationship types can be created with conformant endpoint type combinations
-

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Level 2 — Meaning-Preserving

What is required: Everything in Level 1, plus:

- All 10 Meaning-Preservation Rules (MPR-001 through MPR-010) implemented and enforced
- All 8 Semantic Commitment Contracts (SC-IDENTITY through SC-CONTRADICTION-FREE) fully implementable and machine-evaluated
- Confidence decay enforcement (MPR-005) on all DERIVES_FROM chains
- CONTEXT inheritance mechanism (MPR-006) operational
- Policy evaluation gate before every governed entity mutation (MPR-008)
- SC-CONFIDENCE-FLOOR monitoring and below-floor event emission

Expected Tooling: Contract management subsystem; confidence tracking with decay propagation; context inheritance resolver; policy evaluation engine; conflict resolution procedure executor.

Required Test Assertions:

- T2-001: DERIVES_FROM chain with $\delta=0.95$ produces correct confidence at each hop (MPR-005)
- T2-002: Node with SC-CONFIDENCE-FLOOR emits BelowFloorEvent when confidence drops below floor
- T2-003: Mutation on a POLICY-GOVERNED entity is blocked when it violates the POLICY (MPR-008)
- T2-004: Cross-version node access without schema compatibility check → SchemaVersionConflictError (MPR-010)
- T2-005: SC-IDENTITY prevents SemanticType change after contract activation
- T2-006: CONTEXT inheritance applies correctly to child nodes; override is logged
- T2-007: SC-CONTRADICTION-FREE blocks CONTRADICTS edge creation on protected node

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Level 3 — Full URIB Conformant

What is required: Everything in Level 1 and Level 2, plus:

- Cross-domain Bridge Operations fully implemented: DomainBridgeRecord creation, type mapping, semantic equivalence checks
- INV-009 enforced on all cross-domain edges
- CLAIM contradiction resolution procedure implemented with at least two resolution policies (Newest-Wins, Confidence-Weighted)
- INV-011 halt-and-resolve cycle for contradicting CLAIMs
- SIGNAL TTL management: ephemeral purge scheduler, StaleSignalError emission (INV-012)
- MPR-009 Signal-to-State elevation with threshold validation and confidence floor check
- SC-CROSS-DOMAIN-FIDELITY contract enforcement on bridged nodes
- Semantic Authority AGENT designation and enforcement (MPR-002 type demotion authorization)
- Full temporal lifecycle management: creation, valid scope, expiry, archival

Expected Tooling: Domain bridge service; type mapping registry; equivalence check engine; contradiction resolution orchestrator; ephemeral TTL scheduler; semantic authority access control; temporal lifecycle manager.

Required Test Assertions:

- T3-001: Cross-domain edge without DomainBridgeRecord → REJECTED (INV-009)
- T3-002: Full Bridge Operation produces correct DomainBridgeRecord with type mapping and SC-CROSS-DOMAIN-FIDELITY
- T3-003: CONTRADICTS edge triggers ContradictionHaltError; Confidence-Weighted resolution selects correct winner
- T3-004: SIGNAL TTL expiry triggers purge and StaleSignalError on consuming edges

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- T3-005: Signal elevation without confidence $\geq 0.80 \rightarrow$ PrematureElevationError (MPR-009)
- T3-006: Type demotion without Semantic Authority AGENT $\rightarrow$ TypeDemotionError (MPR-002)
- T3-007: Complete provenance chain reconstruction from a 5-hop DERIVES_FROM chain
- T3-008: SC-PROVENANCE-AUDITABLE blocks deletion of provenance-chain ancestor node

11. Glossary

Claim — A CLAIM entity node representing an asserted proposition with declared confidence and provenance. Subject to contradiction and resolution procedures.

Confidence — A float in $[0.0, 1.0]$ encoding the semantic certainty of a node's payload. Propagates with decay through DERIVES_FROM chains per MPR-005. Enforced in bounds by INV-007.

Domain Bridge — A Bridge Operation that creates a DomainBridgeRecord enabling an edge to cross a Semantic Domain boundary in compliance with INV-009. Requires type mapping, semantic equivalence check, and an active MeaningPreservationContract.

Entity — Any node $v \in V$ in the semantic graph. Every entity has a SemanticType from the normative Entity Taxonomy (Section 4), a typed Payload, a Provenance record, and a MutabilityClass.

Invariant — A runtime-enforced constraint that MUST hold for all states of the semantic graph. Violation of a MUST-level invariant causes the triggering mutation to be atomically rejected. See Section 6.

Meaning-Preservation Rule (MPR) — A behavioral rule governing how semantic fidelity is maintained across graph operations, including transformations, derivations, serializations, and cross-domain transfers. See Section 7.

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MutabilityClass — A declared property of a node or edge that governs whether it may be modified after creation. Values: `Immutable` (never modifiable, INV-008), `Mutable` (modifiable under policy), `Ephemeral` (purged after TTL expiry, INV-012).

Provenance — The origin context of a node, including source system, source timestamp, ingestion method, and originating agent. Mandatory on every node. DERIVES_FROM chains must be traceable to an authoritative root per INV-010.

Relationship — A typed directed edge $e \in E$ in the semantic graph, connecting a source node to a target node with a RelationshipType from the normative Relationship Taxonomy (Section 5). Carries Weight, Directionality, and TemporalScope.

Semantic Authority — A designated AGENT node with a GOVERNS relationship over a Semantic Domain. Semantic Authorities may authorize type demotion (MPR-002), invalidate contracts, and extend temporal scopes. Authority designation is itself governed by a POLICY.

Semantic Commitment Contract (SCC) — A formal machine-evaluable declaration that a node will maintain specified semantic properties across its lifetime. Contracts are evaluated at every relevant runtime operation. See Section 8.

Semantic Domain — A named, namespaced subgraph partition of the URIB Semantic Graph with its own type schema version (Σ_d), governing POLICY nodes, and Semantic Authority. Every node belongs to exactly one domain. Cross-domain edges require a Bridge Operation.

Semantic Graph — The formal structure $G = (V, E, \Sigma, \Phi)$ constituting the URIB truth surface. All semantic tokens are nodes in this graph; all typed relationships between tokens are edges. The graph is the authoritative meaning representation.

Semantic Token — A typed, graph-anchored entity node produced by the Semantic Tokenization process. Carries explicit SemanticType, Payload, Provenance, Confidence, MutabilityClass, and TemporalScope. The unit of meaning in the URIB model.

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Semantic Tokenization — The process of transforming raw data primitives (strings, values, events, references) into typed, graph-anchored semantic tokens. The foundational operation of the URIB semantic layer.

Temporal Scope — The validity window of a node or edge, declared as `valid_from` and `valid_to` in ISO 8601 UTC. Nodes outside their TemporalScope MUST NOT be treated as authoritative for current-state queries.

Type Schema (Σ) — The normative mapping from SemanticType identifiers to payload schemas, allowed relationship sets, and behavioral constraints. Domain-specific schemas are versioned (Σ_{d_v}). Cross-version operations require compatibility checks per MPR-010.

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12. Revision History & Authorship

Version	Date	Status	Description	Notes
1.0.0	July 2026	Authoritative	Initial authoritative release of the URIB Semantic Tokenization Specification.	Supersedes all prior informal specifications and Base44 platform implementation notes. This is the canonical normative reference.

Authorship & Stewardship

This specification is stewarded by the **URIB Semantic Working Group** under the Universal Bridge Framework governance structure. The Semantic Working Group is responsible for:

- Issuing normative amendments and errata to this specification
- Evaluating proposed extensions to the Entity and Relationship Type Taxonomies
- Maintaining the URIB Conformance Test Suite aligned with the conformance levels in Section 10
- Approving schema version increments (Σ updates) that affect cross-domain compatibility

Supersession Notice

Supersession Notice

This document (URIB-SPEC-001 v1.0.0) supersedes and replaces all prior informal specifications, design documents, wiki pages, and Base44 platform implementation notes relating to semantic tokenization, entity typing, or semantic graph construction in URIB systems. Prior documents are retained for historical reference

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only and carry no normative weight. In any conflict between a prior document and this specification, this specification prevails.

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Companion Specifications

This specification is part of the URIB normative document family. Related companion specifications (not yet released):

Document ID	Title	Status
URIB-TRANSPORT-001	URIB Physical Transport Protocol Specification	Planned
URIB-RUNTIME-001	URIB Runtime Orchestration Specification	Planned
URIB-CONFORMANCE-001	URIB Conformance Test Suite and Certification Procedure	Planned
URIB-SCHEMA-001	URIB Type Schema Registry and Versioning Protocol	Planned